CS2201 Computer Organization & Architecture

L35: Hard Disk Organization



John Jose

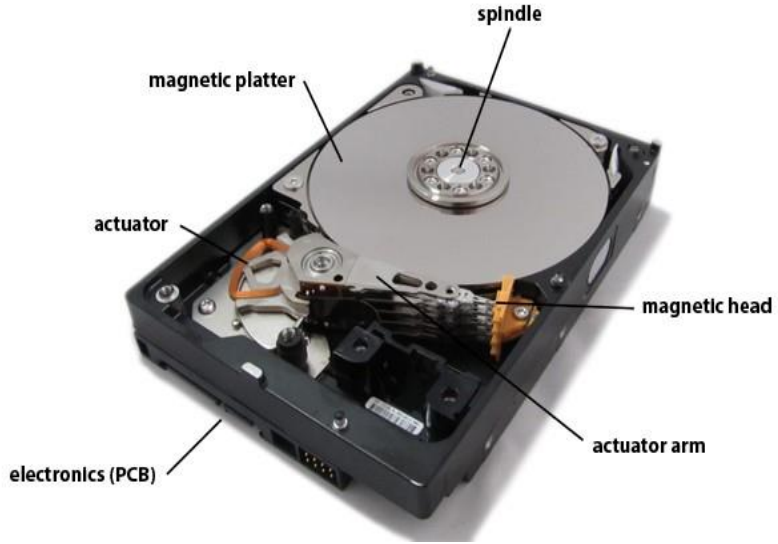
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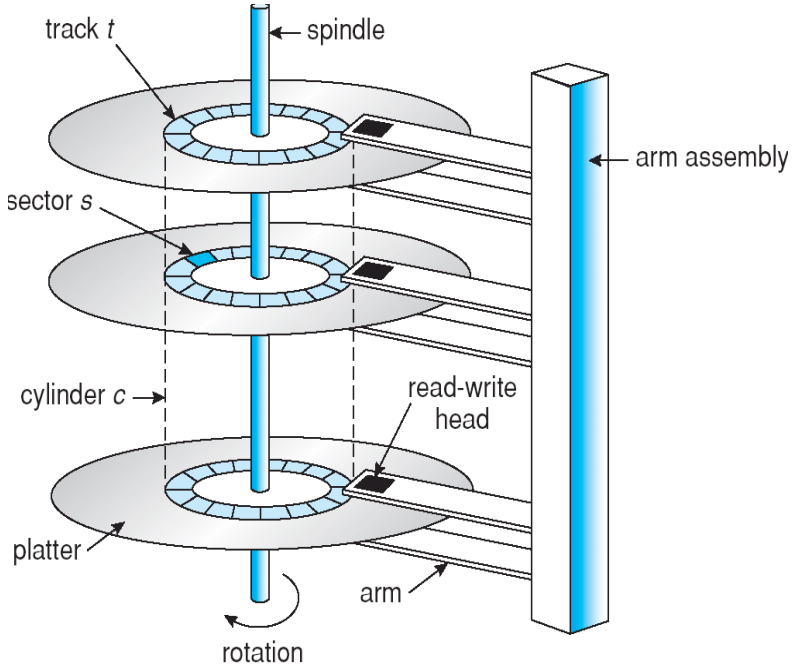
Mass Storage - Hard Disk

- Systems today need to store many terabytes of data.
- Primary level of permanent storage is hard disk.
- Electromechanical
  - Rotating disks
  - Arm assembly
- Electronics
  - Disk controller
  - Cache
  - Interface controller



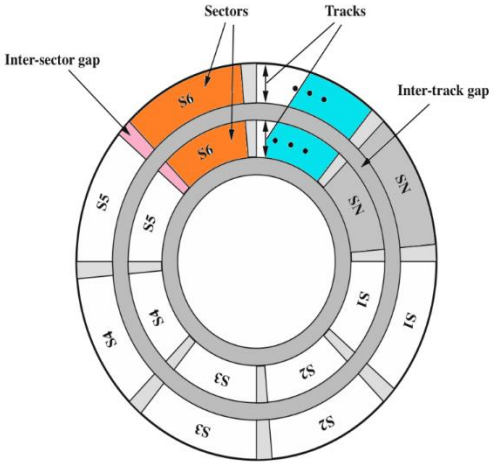
Hard Disk Organization

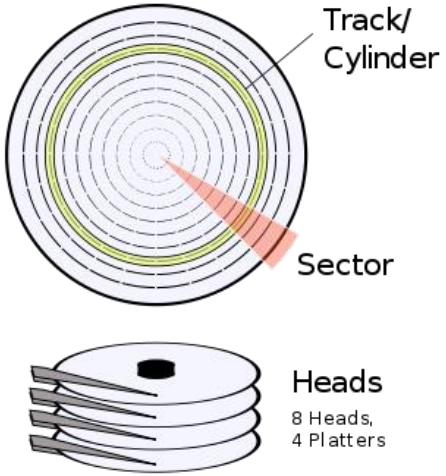
- Spinning disks with heads that move over the disks and store data in tracks and sectors.
- The heads read and write data in concentric rings called tracks.
- Tracks are divided into sectors, which normally store 512 bytes each.



Hard Disk Organization

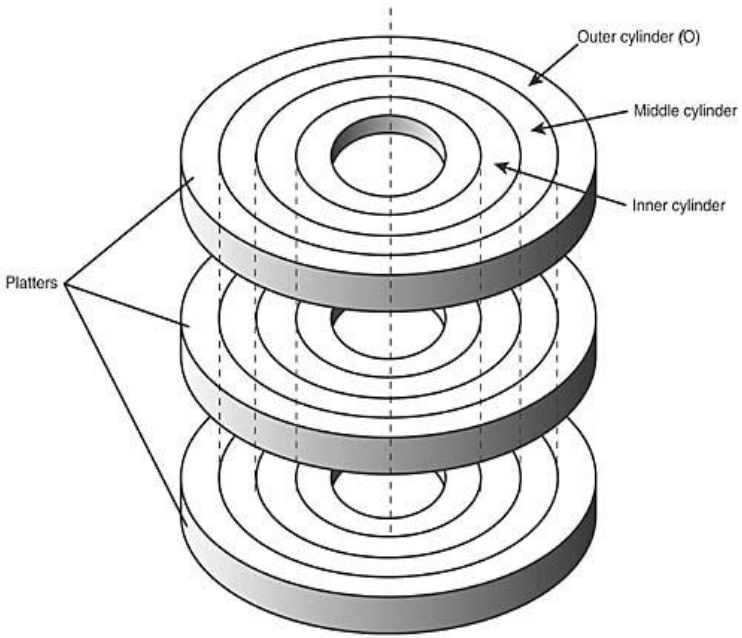
- One side of a platter is called a head.
- Multiple platters, depending on their design and storage capacity.
- There are concentric rings (tracks) and pieces of rings (sectors)





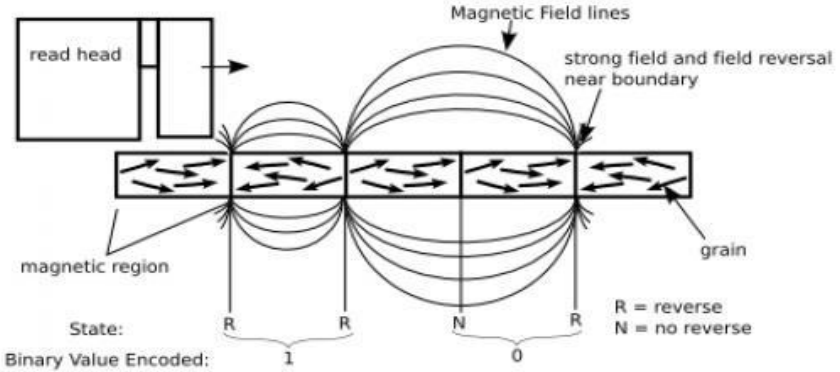
Hard Disk Organization

- Cylinder: 3D collection of track ‘n’of each surface of all platters.
- Sector 0 is the first sector of the first track on the outermost cylinder
- Mapping proceeds in order through that track, then the rest of the tracks in that cylinder, and then through the rest of the cylinders from outermost to innermost



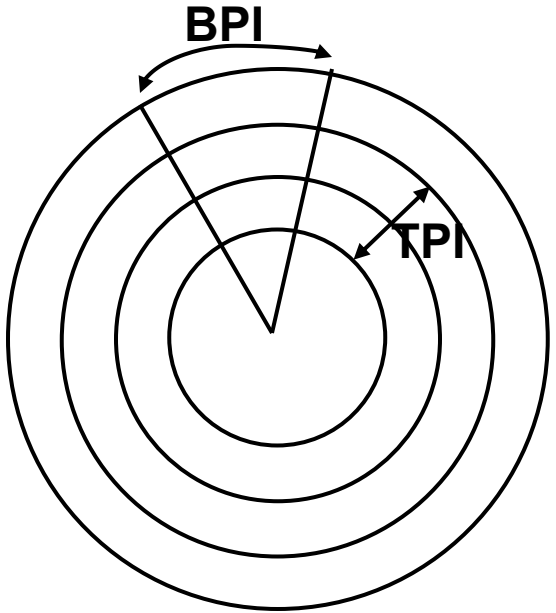
Logic Storage in Hard Disk Drive

- Bit-cell composed of magnetic grains : 50-100 grains/bit
- Size of grains is order of 10 nm.
- **‘0’** Region of grains of uniform magnetic polarity
- **‘1’** Boundary between regions of opposite magnetization
- The read-and-write head is used to detect and modify the magnetization of the material immediately under it.



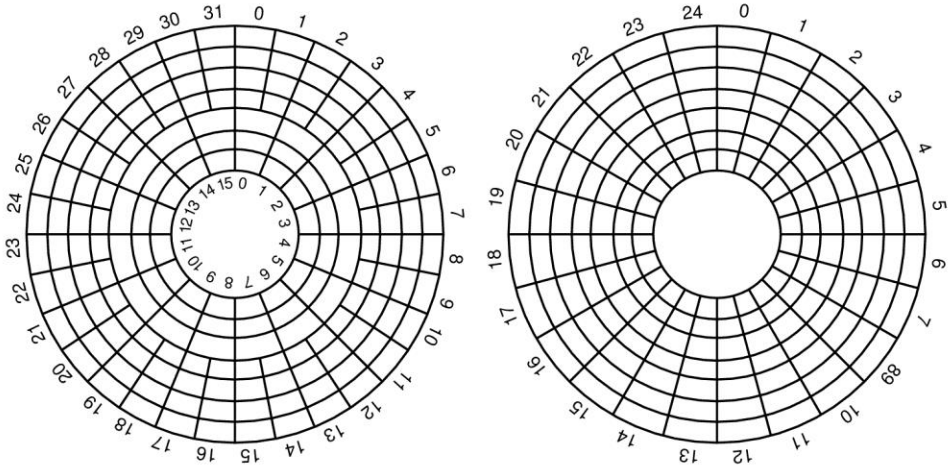
Storage Density

- Determines both capacity and performance
- Density Metrics
  - Linear density (Bits/inch or BPI)
  - Track density (Tracks/inch or TPI)
  - Areal Density = BPI x TPI



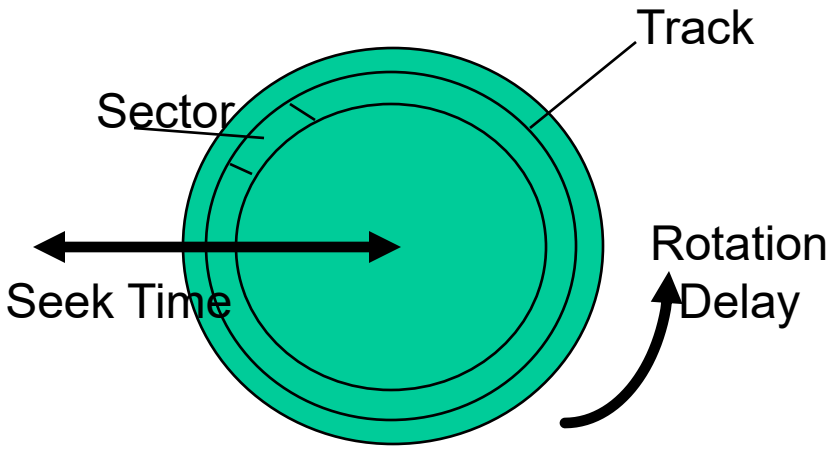
HDD- Bit Density

- Reduce bit density per track for outer layers. Constant Linear Velocity.
- Have more sectors per track on the outer layers, and increase rotational speed when reading from outer tracks. Constant Angular Velocity.



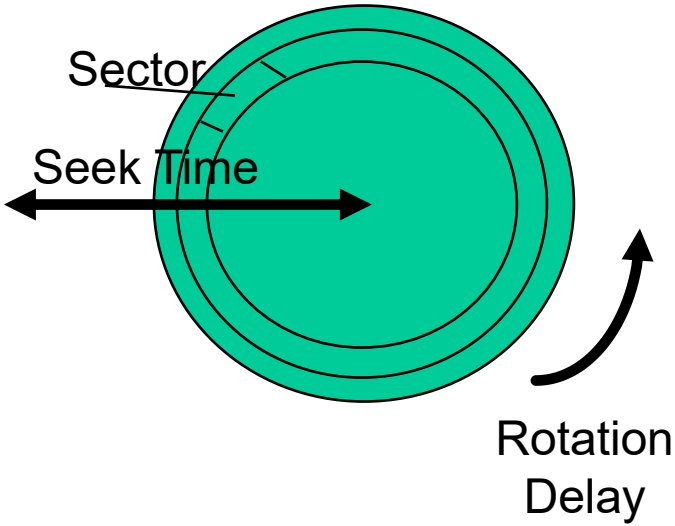
Disk Access Time

- To read from disk, we must specify:
  - cylinder #, surface #, sector #, size, memory address
- Transfer time includes:
  - Seek time: to get to the track
  - Rotational Latency: to get to the sector
  - Transfer time: get bits off the disk



Seek Time

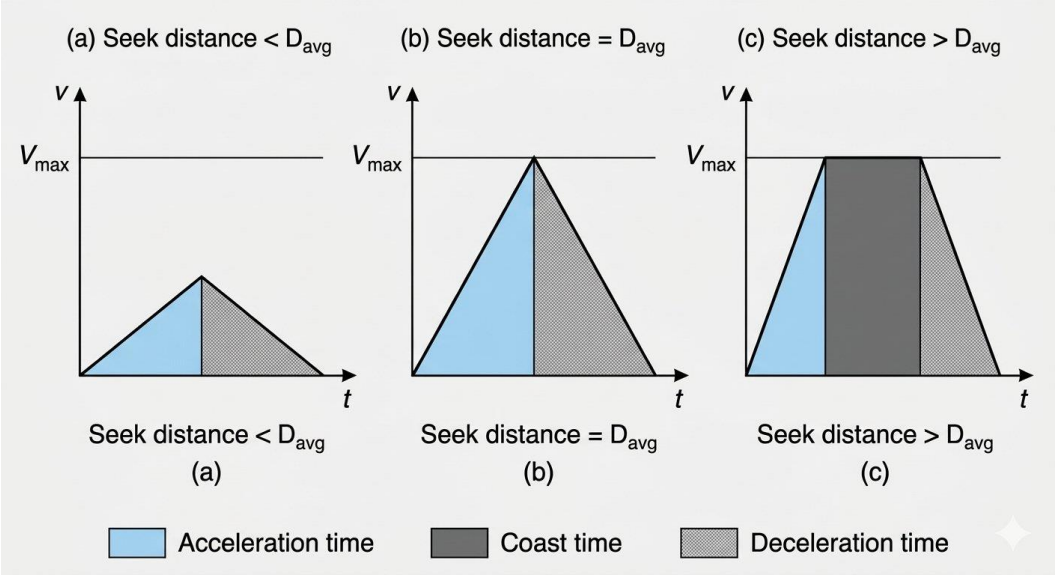
- **Seek time depends on:**
  - Inertial power of the arm actuator motor
  - Distance between outer-disk recording radius and inner-disk recording radius (data-band)
  - Depends on platter-size



- **Components of a seek:**
  - Speedup: Arm accelerates
  - Coast: Arm moving at maximum velocity (long seeks)
  - Slowdown: Arm brought to rest near desired track
  - Settle: Head is adjusted to reach the access the desired location

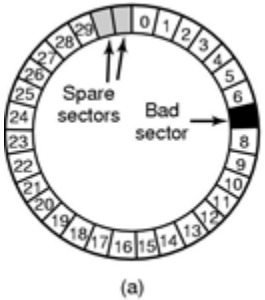
Variations in Seek Time

- Very short seeks (2-4 cylinders) - Settle-time dominates
- Short seeks (100-200 cylinders) - Speedup/Slowdown-time dominates
- Longer seeks (> 200 cylinders) - Coast-time dominates

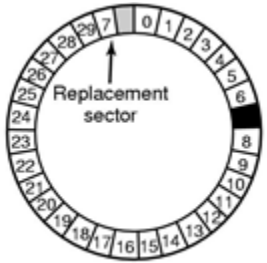


Bad Sector Management in Disks

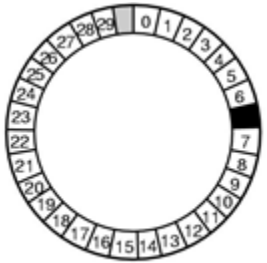
- **Bad sector** is a sector on disk that is either inaccessible or un-writeable due to permanent damage.
- Bad sectors are usually detected by formatting or by utility software such as CHKDSK or SCANDISK .
- The sectors unusable are not used for storage.
- If a file uses a sector which is marked as bad then the bad sector of the file is remapped to a free sector.



(a)

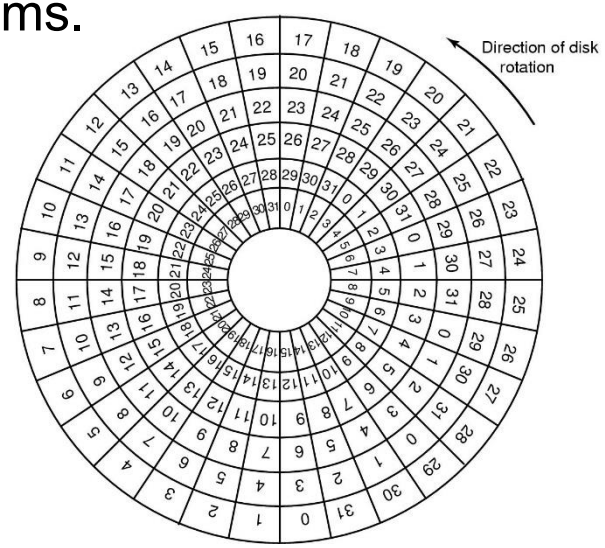


Sector forwarding


Sector slipping

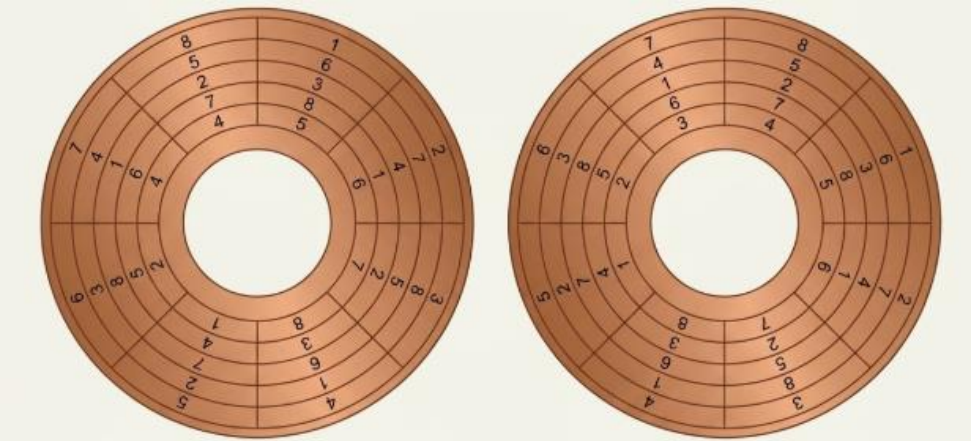
Cylinder Skew

- Offsetting the start sector of adjacent tracks to minimize the likely wait time (rotational latency) when switching tracks
- Example, if 10000 rpm disk drive rotates in 6 ms.
  - Track has 300 sectors
  - New sector every 20 μs
  - If track seek time 800 μs
    - 40 sectors pass on seek
- Cylinder skew: 40 sectors



Head Skew

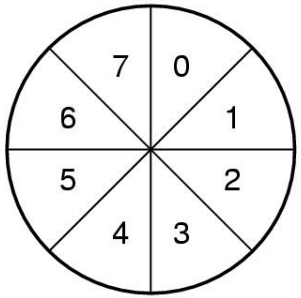
- Head skew is the offsetting done on the start sector of tracks of adjacent platters (heads) of same cylinder.
- During head switch, there is no physical movement of arm assembly. But it still takes time for the switch from reading one head to reading another.



Staggering of sector start points on a given platter (left) illustrates CYLINDER SKEW; the difference in start points from platter-to-platter (comparing left and right) illustrates HEAD SKEW.

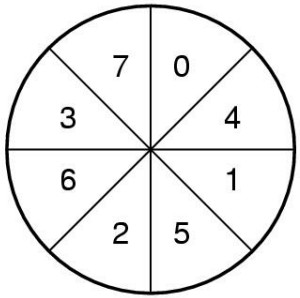
Sector Interleaving

- To ensure that sector #n+1 didn't rotate past the head while sector #n was being processed.



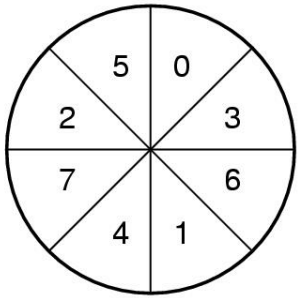
(a)

No interleaving



(b)

Single interleaving



(c)

Double interleaving



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